Every name I've used so far, and what it is

Colored tag = which kind of thing it is. You don't need to memorize these. They're the "for example" behind each category, so when a name shows up you know what bucket it belongs in.

LIVE **Laika** — a Moscow stray dog, the first animal to orbit Earth, inside Sputnik 2 (1957). Proved a living body survives launch and weightlessness. Didn't survive the flight.

LIVE **ISS** — International Space Station. A football-field-sized lab in LEO (~400 km) where 6–7 astronauts live for months. Built by the US, Russia, Europe, Japan, Canada. Orbits every 90 minutes.

LOOK OUT **Hubble** — a bus-sized telescope orbiting Earth since 1990 at ~540 km. Above the air, so razor-sharp pictures. Source of most famous deep-space photos.

LOOK OUT **Kepler** — a space telescope (2009–2018) that stared at one patch of sky and found thousands of planets around other stars.

LOOK DOWN **GOES** — US weather satellites in GEO. The cloud loops on the TV weather report are GOES pictures.

RELAY **Starlink** — SpaceX's internet constellation: thousands of small satellites in LEO (~550 km) relaying internet to dishes on the ground.

FLYBY **Voyager 1 & 2** — twin NASA probes launched 1977. Flew past Jupiter and Saturn (Voyager 2 also Uranus and Neptune), then kept going. Both have now left the solar system and are still transmitting.

LIVE **Sputnik 2** — the Soviet satellite that carried Laika. Sputnik 1, a month earlier, was the very first artificial satellite of any kind: a beeping metal ball.

LIVE **Tiangong** — China's own space station, smaller than ISS, also in LEO.

LOOK OUT **JWST** — James Webb Space Telescope (2021). Hubble's bigger successor, sees infrared (heat-light). Parked 1.5 million km from Earth at a gravity balance point, not in Earth orbit.

LOOK DOWN **Landsat** — US series of Earth-mapping satellites running since 1972. Farms, forests, cities, floods, all photographed from LEO.

RELAY **GPS** — Global Positioning System: ~30 US satellites in MEO (~20,200 km) broadcasting time signals. Your phone hears several and computes where you are. Europe's version is Galileo.

RELAY **Queqiao** — a Chinese relay satellite parked beyond the Moon so a lander on the Moon's far side (Chang'e-4) could talk to Earth, since the Moon blocks direct radio.

FLYBY **New Horizons** — NASA probe launched 2006, flew past Pluto July 2015 after 9.5 years, took the famous heart-shaped-glacier photos, then continued into the Kuiper Belt.

Orbiters, probes, landers, rovers, and the places

Second half. Places at the bottom, because a few of them (Titan, Pluto, Kuiper Belt) came up as destinations.

ORBITER **Cassini** — NASA spacecraft that orbited Saturn 2004–2017, photographing the rings and moons. Deliberately crashed into Saturn at the end so it couldn't contaminate a moon.

ATMOSPHERIC **Galileo probe** — dropped by the Galileo orbiter into Jupiter's atmosphere in 1995. Transmitted for 58 minutes before heat and pressure crushed it. No surface to land on.

ORBITER **MRO** — Mars Reconnaissance Orbiter (2006–). Maps Mars in high detail AND relays radio for the rovers on the ground. Orbiter and relay at once.

LANDER **InSight** — NASA Mars lander (2018–2022). Planted a seismometer and listened for marsquakes. Died when dust covered its solar panels.

ROVER **Sojourner** — the first Mars rover (1997), microwave-oven sized.

ROVER **Curiosity & Perseverance** — car-sized, nuclear-powered Mars rovers (2012 and 2021). Perseverance carried the Ingenuity helicopter and is collecting samples for a future return.

PENETRATOR **LCROSS** — NASA mission (2009) that crashed a spent rocket stage into a dark Moon crater and flew through the dust plume. Found water ice.

Pluto — dwarf planet at the edge of the solar system, ~33 times farther from the Sun than Earth. Visited once, by New Horizons.

L2 — a gravity balance point 1.5 million km behind Earth (away from the Sun) where a spacecraft can hover with little fuel. JWST lives there.

Walter Hohmann / Johann Lambert — not missions, people. Hohmann: German engineer who worked out the cheapest two-burn transfer in 1925. Lambert: 18th-century mathematician whose "two points plus a time" orbit problem still carries his name.

GLONASS / BeiDou — Russia's and China's navigation constellations, both in MEO like GPS. Your phone usually listens to all of them.

ATMOSPHERIC **Huygens** — a European probe Cassini carried and dropped onto Titan (Saturn's biggest moon) in 2005. Parachuted through the orange haze, landed, sent pictures for 90 minutes. One mission, two types.

ORBITER **Juno** — NASA Jupiter orbiter, arrived 2016. The huge solar arrays in your clean-room slide are Juno's; it's the farthest solar-powered spacecraft ever.

LANDER **Viking 1 & 2** — NASA's first successful Mars landers (1976). Sat in one spot, photographed the surface, tested the soil for life.

LANDER **Rosetta / Philae** — Rosetta = European spacecraft that orbited a comet (2014). Philae = the small lander it dropped onto the comet's surface. Philae bounced twice, landed in a shadow, ran out of solar power in 3 days.

ROVER **Spirit & Opportunity** — twin golf-cart-sized Mars rovers (2004). Designed for 90 days; Opportunity drove for 14 years. The rover in your slide is this design.

PENETRATOR **Deep Impact** — NASA mission (2005) that fired a washing-machine-sized copper block into comet Tempel 1 and watched what sprayed out.

Titan — Saturn's largest moon. Thick orange atmosphere, lakes of liquid methane. The only moon anyone has landed a probe on (Huygens).

Kuiper Belt — the ring of icy leftovers beyond Neptune. Pluto is its biggest known member.

Comet 67P / Churyumov–Gerasimenko — the duck-shaped comet Rosetta orbited from 2014 and Philae landed on. Named after the two astronomers who spotted it in 1969.

Galileo (two things) — the European navigation constellation (GPS rival, MEO), AND NASA's 1990s Jupiter orbiter that dropped the atmospheric probe. Context tells you which.

NASA / ESA / SpaceX — US space agency / European Space Agency / the private US rocket company (Falcon rockets, Starlink). Not affiliated with your notebook title.

Section 4: rockets, engines, and the people behind them

Every proper name used in the Section 4 propulsion PDFs so far. Same rule as before: if I use a name, it is defined here.

ROCKET **Saturn V** — the Apollo Moon rocket (flew 1967–1973). 110 m tall, three stages. Its first stage had five F-1 engines and burned 13 tonnes of propellant every second. The rocket lying on its side in your slide.

ROCKET **Space Shuttle / SLS** — NASA's reusable spaceplane (1981–2011) and its current Moon rocket. Both use RS-25 hydrogen engines plus two big solid boosters.

ENGINE **Merlin / Merlin Vacuum** — SpaceX's kerosene + liquid oxygen engine. Nine on Falcon 9's first stage; one 'Vacuum' version with a huge bell on the second stage (981,000 N, Isp 348 s).

ENGINE **NEXT (and NSTAR)** — NASA's ion thrusters. NSTAR flew on Deep Space 1 and Dawn; NEXT, the blue glow in your slide, is the bigger successor (0.236 N, Isp 4,190 s) and flew on DART.

PROBE **Dawn** — NASA ion-engine probe (2007–2018). Orbited the asteroid Vesta, left, then orbited the dwarf planet Ceres: the first spacecraft to orbit two different bodies, only possible with electric propulsion.

PROBE **BepiColombo / Psyche** — European–Japanese Mercury mission (launched 2018, ion engines) / NASA mission to a metal asteroid (launched 2023, Hall thrusters). Today's electric-propulsion flagships.

PROGRAM **NERVA** — the US nuclear thermal rocket program. Built and ground-tested working engines in Nevada in the 1960s; ended in 1972 without ever flying.

PROGRAM **Project Rover** — the US reactor-testing effort (from 1955) that fed into NERVA. About 20 nuclear rocket reactors were run in Nevada between 1959 and 1972.

CAPSULE **Orion (crew capsule)** — NASA's current astronaut capsule for Moon missions. Shares only the name with Project Orion.

THING **RTG** — radioisotope thermoelectric generator: a plutonium power pack that turns decay heat into electricity. Powers Voyager, New Horizons, Curiosity. Not an engine.

PERSON **Konstantin Tsiolkovsky** — Russian schoolteacher who published the rocket equation in 1903, before any liquid rocket existed.

PERSON **Gustaf de Laval** — Swedish engineer who invented the narrowing-then-widening nozzle in 1888, for steam turbines. Every rocket bell is one.

ROCKET **Falcon 9** — SpaceX's two-stage workhorse rocket. The first stage flies back and lands; the second stage, the one in your slide, has a single Merlin Vacuum engine.

ENGINE **F-1** — the Saturn V first-stage engine: kerosene + liquid oxygen, 6.77 million newtons each. Still the most powerful single-chamber liquid engine ever flown.

ENGINE **RS-25** — the Space Shuttle main engine: liquid hydrogen + liquid oxygen, Isp 452 s. About the best mileage a chemical engine gets.

PROBE **Deep Space 1** — NASA technology test (1998). The first probe to cruise between planets on an ion engine.

PROBE **DART** — NASA mission that deliberately hit a small asteroid moon in 2022 to test deflecting one. Carried a NEXT ion thruster.

TEST **SERT-1 / Zond 2** — the first electric thrusters fired in space, both in 1964: a NASA ion-engine test and a Soviet Mars probe's plasma thrusters. Electric propulsion is 60 years old.

PROGRAM **DRACO** — a NASA and DARPA plan to fly a nuclear thermal rocket in orbit. Cancelled in 2025, as far as I know.

PROGRAM **Project Orion (1958–1965)** — a US study of nuclear PULSE propulsion: a ship riding small nuclear explosions on a pusher plate. Never built; ended by the 1963 treaty banning nuclear explosions in space. Not a nuclear thermal rocket, and not the Orion crew capsule.

PLACE **Proxima Centauri** — the nearest star to the Sun, 4.2 light years away: about 40 trillion km. At Voyager's speed, 75,000 years.

PERSON **Isaac Newton** — English scientist whose three laws of motion (1687) are the whole basis of rocketry. The third: every action has an equal and opposite reaction.

PERSON **Robert Goddard** — American who flew the first liquid-fuel rocket in 1926. A 1920 newspaper editorial mocked him for thinking rockets could work in vacuum; the paper corrected itself in 1969, as Apollo 11 flew to the Moon.

Section 5: radio, antennas, and who listens

Every proper name used in the Section 5 communications PDFs so far. More will be added as the section goes on.

PERSON **James Clerk Maxwell** — Scottish physicist who showed in the 1860s that electricity and magnetism together make waves that travel at the speed of light. Every radio link rests on his equations.

THING **CubeSat** — a tiny satellite built from 10 cm cubes, cheap enough for universities. The one in your slide has four thin rod antennas that spring out after launch.

MISSION **Apollo 17** — the last crewed Moon landing, December 1972. Two astronauts spent three days on the surface.

NETWORK **Deep Space Network (DSN)** — NASA's three ground stations in California, Spain and Australia, spaced so one always faces any spacecraft. Each has a 70-metre dish.

AGENCY **ITU** — the International Telecommunication Union, the United Nations body that decides which radio frequencies may be used for what. It is why GPS, Wi-Fi and a probe at Saturn do not jam each other.

PERSON **Ludwig Boltzmann** — Austrian physicist (1844–1906) who connected temperature to the random motion of particles. The constant named after him, k, turns a receiver's temperature into its noise power.

PERSON **Heinrich Hertz** — German physicist who first made and detected radio waves in 1887, proving Maxwell right. The unit of frequency is named after him.

ORBITER **Cassini (its radio)** — the Saturn orbiter from page 2. Its 4-metre dish was its high-gain antenna; the transmitter behind it was only 20 watts.

VEHICLE **Lunar Roving Vehicle** — the battery-powered car driven on the Moon on Apollo 15, 16 and 17. It carried a wide-beam low-gain antenna for voice while driving and an umbrella-shaped high-gain dish for television when parked.

THING **Yagi antenna** — the fishbone-shaped rooftop TV antenna, named after one of its two Japanese inventors (1926). A simple directional antenna.

THING **Loran-C** — a long-range radio navigation system that used low frequencies (around 100 kHz) before GPS replaced it. Named on your frequency slide.

PROBE **Voyager 1 (its radio)** — the farthest spacecraft, launched 1977. Its 23-watt transmitter now reaches Earth at 160 bits per second.